1. Introduction

The Study Room Management System is a web-based platform designed to efficiently manage and organize shared study spaces, also known as study rooms, within university halls. Students often encounter difficulties in finding available seats, handling unauthorized seat occupation, and managing study time slots in an organized manner. This system enables students to book seats in advance, check real-time seat availability, and manage their reservations digitally. At the same time, administrators can monitor overall usage, enforce rules, and ensure fair allocation of study spaces. The platform aims to establish a structured, transparent, and efficient study environment that improves resource utilization and reduces conflicts among users.

2. Motivation

Our university halls face very common but solvable problems. These are described below:

i) In university halls, most study room seats appear occupied even when many are not actively in use, creating an unnecessary shortage of available space for other students.

ii) A common issue is that some students occupy multiple seats, especially during peak hours such as DL time, leading to unfair usage and increased difficulty for others to find seating.

iii) Certain areas like the relaxation corner are often “reserved” or claimed without actual presence, further reducing the availability of genuine study space.

iv) Students who do not reside in the hall often stay in the study room during off-hours or breaks. However, even they struggle to find seats.

v) The small library beside the study room suffers from irregular access and poor management, as it frequently remains closed, preventing access to books and resources.

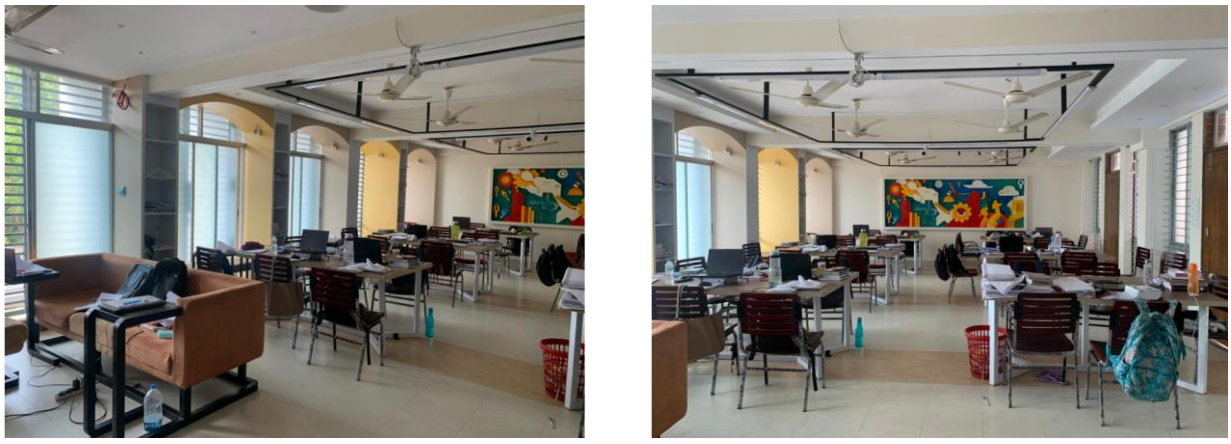


Figure 01: Study room with no students but all seats occupied (Shamshen Nahar Khan Hall)

3. Application

Our platform, StudyTime, provides several practical and impactful solutions for managing study room activities efficiently. It allows students to reserve available study room seats for specific time slots in advance, ensuring fair and organized seat allocation. Users can also update their booking details or cancel reservations when needed, improving flexibility and efficient seat utilization. The platform includes a structured complaint system through which students can report issues such as unauthorized seat occupation, misuse of seats, or other management problems that can be reviewed by administrators. This system also provides scope for improving the platform based on user feedback. In addition, the platform facilitates access to the small library beside the study room by creating a digital lending system for books. Students can check real-time seat availability to see which seats are free or occupied, reducing confusion and saving time while searching for study space, while also removing the need to pre-occupy seats. Furthermore, the system assists administrators in monitoring seat usage patterns and ensuring that no student occupies multiple seats or misuses designated study areas.

4. Background Study

We reviewed some existing real-world systems related to seat booking, space management, and resource sharing to understand their features and limitations. These systems are widely used, but none are specifically designed for managing university hall study rooms.

Skedda – Space Booking System [1]

Skedda is an online space booking platform used by universities and organizations to manage rooms, desks, and shared spaces.

Limitations:

• Designed for offices and institutions, not specifically for student hall environments

• Requires full organizational setup and configuration

• Limited customization for student-specific issues like seat misuse or complaints

• Often used for formal workspace booking, not informal study room usage

Google Calendar / Google Sheets Booking System [2]

Some institutions use Google Calendar or Sheets for simple reservation tracking.

Limitations:

• No real-time seat validation

• Easy to edit or manipulate entries

• No authentication or strict access control

• Not suitable for managing physical seat occupation issues

• Lacks automated conflict resolution

• Not customized for the space in use

LibCal (Library Seat Booking System) [4]

LibCal is a library space management system used in universities for booking study rooms and library seats.

Limitations:

• Primarily designed for libraries and formal study room environments.

• Requires institutional subscriptions and additional setup costs.

• Lacks flexibility in managing informal hall-based seating arrangements.

• Does not effectively prevent misuse scenarios, such as occupying multiple seats at once.

5. Potential Improvements

Real-Time Seat Validation

StudyTime provides live seat availability checking, ensuring only actually free seats can be booked, unlike manual systems such as Google Sheets or Calendar.

Secure Authentication

The system uses Student ID and contact information for login, ensuring only verified users can access the platform and improving overall security.

Conflict Prevention

Automated booking control prevents multiple users from reserving the same seat at the same time, avoiding scheduling conflicts.

Hall-Specific Customization

Unlike general systems like Skedda or LibCal, StudyTime is specifically designed for university hall study rooms, supporting DL/PL priority seating and informal usage patterns.

Complaint Management System

Students can report issues such as seat misuse or facility problems, allowing administrators to take quick action and maintain discipline.

Visual Seat and Calendar System

A color-coded seat map and calendar-based scheduling make seat selection simple, clear, and user-friendly.

Fair Usage Monitoring

The system helps administrators track usage patterns and prevent misuse such as occupying multiple seats or unfair booking behaviour.

6. Description

The proposed StudyTime system is designed as an integrated solution to manage a single university hall study room in a structured and efficient manner. It brings together multiple functional components into a unified database system, enabling better coordination between students, seat reservations, complaints, and library activities.

A key component of the system is the Student Management module, which stores detailed information about students, including Student ID, Name, Phone Number, Hall Name, Room Number, Email, and Password. The system uses this information for authentication and to uniquely identify each student within the platform.

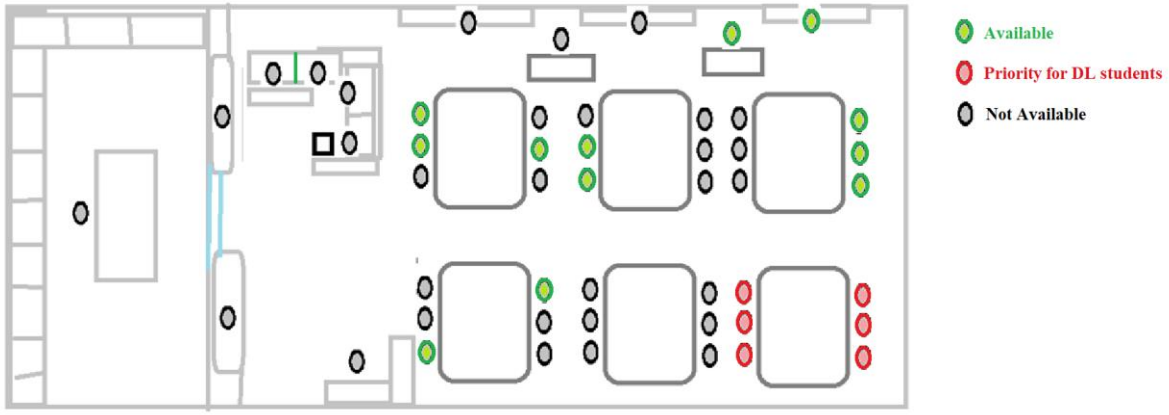


Figure 02: Color-coded seat map : Green (Available), Red (Priority for DL students) and Grey (Unavailable).

The system includes a Seat Management module, where each seat in the study room is stored with details such as Seat Number, Seat Status (Available, Reserved, or Unavailable), and Priority Type (General or DL/PL Priority). This module ensures proper tracking of all seats in the study room and supports fair allocation.

Through the Seat Reservation module, students can book seats by selecting a Seat Number, Date, and Time Slot. Each reservation is recorded with Reservation ID, Student ID, Seat Number,

Booking Date, Time Slot, and Status. The system allows students to create, update, and cancel reservations, ensuring flexible and efficient seat usage.

The system includes a Calendar and Availability module that displays all reservations in a structured schedule format. This allows students to check real-time seat availability before making a booking and helps reduce conflicts in seat selection.

The system also includes a Complaint Management module, where students can submit complaints related to seat misuse, unauthorized occupation, facility issues, or general problems. Each complaint is stored with Complaint ID, Student ID, Complaint Type, Description, Submission Date, and Status. Administrators are responsible for reviewing and resolving these complaints.

The Library Management module handles books available in the study room library. Each book is stored with Book ID, Book Name, Author Name, Category, Availability Status, Borrow Date, and Return Date. Students can borrow and return books, while the system keeps track of all borrowing activities and availability.

The system includes an Administrator Management module, where administrator details such as Admin ID, Name, Phone Number, Email, and Password are stored. Administrators manage students, seat allocation, reservations, complaints, and library operations to ensure smooth system functioning.

Additionally, the system supports Real-Time Seat Monitoring, allowing students to view seat availability instantly through a visual interface. This ensures transparency and reduces unnecessary seat blocking or confusion.

Overall, StudyTime provides a structured and centralized approach to managing study room operations, ensuring fairness, transparency, and efficient resource utilization within a single study room environment.

7. Conclusion

The Study Room Management System is a student-focused initiative aimed at improving the management and utilization of shared study spaces within university halls. It addresses key issues such as uncertainty in seat availability, unfair occupation of seats, and the absence of a proper monitoring mechanism, all of which negatively affect students' ability to access study facilities efficiently.

The proposed system introduces a structured approach to managing study room resources by enabling seat availability tracking, advance booking, and systematic reservation handling. These features are intended to promote transparency, ensure fair usage, and reduce conflicts among students regarding space allocation.

Overall, the Study Room Management System aims to enhance discipline, fairness, and efficiency in the use of study spaces, thereby contributing to a more organized and productive academic environment within the university.

References

[1] Skedda. https://www.skedda.com

[2] Google Calendar. https://calendar.google.com

[3] Google Sheets. https://www.google.com/sheets/about

[4] LibCal. https://www.springshare.com/libcal